

DDA5002 Optimization

Lecture 10 Optimality Conditions

KKT Conditions

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Announcement

- Homework 4 is due this Sunday (Nov 16).
- Final exam will be on Monday, December 15, from 10:00am to 12:00pm.

Last lecture : Dec 10

Outline

- 1 Optimality Conditions for Unconstrained Problems
- 2 Linear Constrained Problems
- 3 Lagrangian Relaxation
- 4 KKT Conditions

Optimality Conditions for Minimization

We consider the unconstrained minimization problem:

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

Theorem (FONC for Minimization)

If $\mathbf{x}^*$ is a local minimizer of f for the unconstrained problem, then we must have $\nabla f(\mathbf{x}^*) = 0$.

Theorem (SONC for Minimization)

If $\mathbf{x}^*$ is a local minimizer of f for the unconstrained problem, then we must have i) $\nabla f(\mathbf{x}^*) = 0$; ii) $\nabla^2 f(\mathbf{x}^*)$ is positive semi-definite. $\nabla^2 f(\mathbf{x}) \succeq 0$

Theorem (SOSC for Minimization)

Let f be twice continuous differentiable. If $\mathbf{x}^*$ satisfies i) $\nabla f(\mathbf{x}^*) = 0$; ii) $\nabla^2 f(\mathbf{x}^*)$ is positive definite, then $\mathbf{x}^*$ is a local minimizer. $\nabla^2 f(\mathbf{x}) \succ 0$

Optimality Conditions for Maximization

OWe consider the unconstrained minimization problem:

$$\max_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

Theorem (FONC for Maximization)

If $\mathbf{x}^*$ is a local maximizer of f for the unconstrained problem, then we must have $\nabla f(\mathbf{x}^*) = 0$.

Theorem (SONC for Maximization)

If $\mathbf{x}^*$ is a local maximizer of f for the unconstrained problem, then we must have i) $\nabla f(\mathbf{x}^*) = 0$; ii) $\nabla^2 f(\mathbf{x}^*)$ is negative semi-definite $\nabla^2 f(\mathbf{x}^*) \preceq 0$

Theorem (SOSC for Maximization)

Let f be twice continuous differentiable. If $\mathbf{x}^*$ satisfies i) $\nabla f(\mathbf{x}^*) = 0$; ii) $\nabla^2 f(\mathbf{x}^*)$ is negative definite, then $\mathbf{x}^*$ is a local maximizer $\nabla^2 f(\mathbf{x}^*) \prec 0$

Critical and Saddle Points



critical Pt $\leftarrow$ local minimizer/maximizer
saddle Pt

Definition

- A point $\mathbf{x}$ satisfying $\nabla f(\mathbf{x}) = 0$ is called **critical point** or **stationary point**.
- A stationary point is called **saddle point** if it is neither a local minimizer nor a local maximizer.

Theorem

Suppose that $\mathbf{x}^*$ is a stationary point ($\nabla f(\mathbf{x}^*) = \mathbf{0}$) and that the Hessian $\nabla^2 f(\mathbf{x}^*)$ is indefinite, then $\mathbf{x}^*$ is a **saddle point**.

$\hookrightarrow$ Positive and negative eigenvalues

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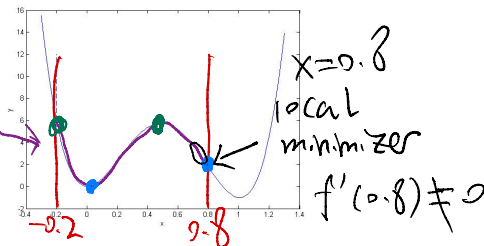
Constrained Problems

We have derived necessary and sufficient conditions for the local minimum for unconstrained problems.

- What is the difference between constrained and unconstrained problems?

Consider the example $f(x) = 100x^2(1-x)^2 - x$ with constraint

$-0.2 \leq x \leq 0.8$.



In addition to the original local minimizer ($x_1 = 0.013$), there is one more local minimizer on the boundary ($x = 0.8$).

Constrained Problems

At the boundary ($x^* = 0.8$), the FONC is not satisfied

$$f'(0.8) < 0$$

However, at this point, in order to stay feasible, we can only go leftward. That is, in the Taylor expansion

$$\rightarrow f(x^* + \textcircled{d}) = f(x^*) + df'(x^*) + o(d)$$

we can only take d to be negative (otherwise it won't be feasible). It means that the developed FONC ($\nabla f(x) = 0$) for unconstrained problem is not enough for constrained problems.

Feasible Directions

Now we formalize the above arguments.



Definition (Feasible Direction)

Given $\mathbf{x} \in F$, we call $\mathbf{d}$ to be a *feasible direction* at $\mathbf{x}$ if there exists $\bar{\alpha} > 0$ such that $\mathbf{x} + \alpha \mathbf{d} \in F$ for all $0 \leq \alpha \leq \bar{\alpha}$.

step size same $\mathbf{d}$

For example,

- If $F = \{\mathbf{x} | A\mathbf{x} = \mathbf{b}\}$, then the feasible directions at $\mathbf{x}$ is $\{\mathbf{d} | A\mathbf{d} = 0\}$
- If $F = \{\mathbf{x} | A\mathbf{x} \geq \mathbf{b}\}$, then the feasible directions at $\mathbf{x}$ is $\{\mathbf{d} | \mathbf{a}_i^T \mathbf{d} \geq 0 \text{ if } \mathbf{a}_i^T \mathbf{x} = b_i\}$

FONC for Constrained Problems

Theorem (FONC for Constrained Problems)

If $\mathbf{x}^*$ is a local minimum of $\min_{\mathbf{x} \in F} f(\mathbf{x})$, then for any feasible direction $\mathbf{d}$ at $\mathbf{x}^*$, we must have $\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0$.

- In unconstrained problems, all directions are feasible, thus we must have $\nabla f(\mathbf{x}^*) = 0$.
- Issue: This FONC is hard to use since we need to consider all feasible directions.

First order Taylor:

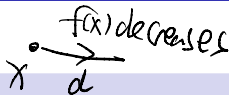
$$f(\mathbf{x}^* + \alpha \mathbf{d}) = f(\mathbf{x}^*) + \alpha \nabla f(\mathbf{x}^*)^T \mathbf{d} + \dots$$

large

small

$$\Rightarrow \alpha \nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0 \Rightarrow \nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0, \text{ for feasible } \mathbf{d}$$

An Alternative View



Definition (Descent Direction)

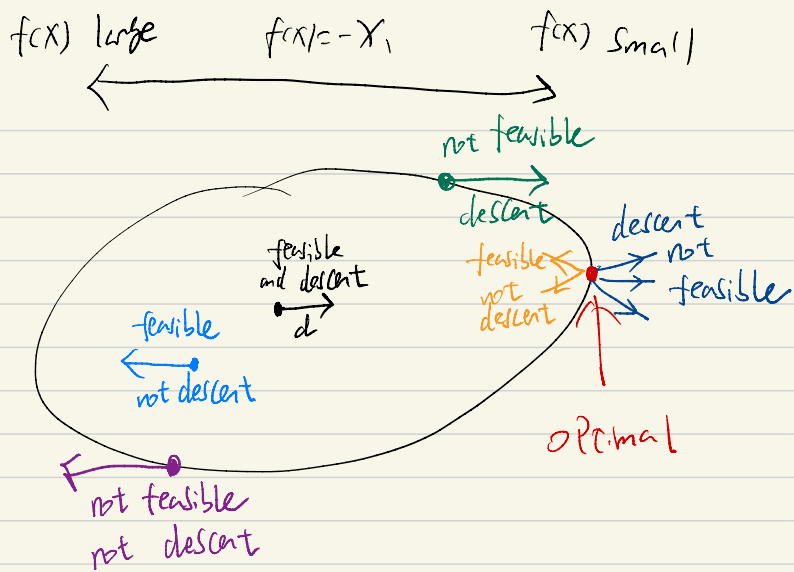
Let f be continuously differentiable. Then $\mathbf{d}$ is called a *descent direction* at $\mathbf{x}$ if and only if $\nabla f(\mathbf{x})^T \mathbf{d} < 0$.

- If $\mathbf{d}$ is a descent direction at $\mathbf{x}$, then there exists $\bar{\gamma} > 0$ such that $f(\mathbf{x} + \gamma \mathbf{d}) < f(\mathbf{x})$ for all $0 < \gamma \leq \bar{\gamma}$.
some γ
- If we denote the set of feasible directions at $\mathbf{x}$ by $S_F(\mathbf{x})$ and the set of descent directions at $\mathbf{x}$ by $S_D(\mathbf{x})$. Then the first order necessary condition can be written as:

$$\rightarrow S_F(\mathbf{x}^*) \cap S_D(\mathbf{x}^*) = \emptyset$$

Or in other words, there cannot be any feasible descent directions.

feasible *descent*



At optimal soln (minimizer), no direction is both feasible and descent.

Nonlinear Optimization with Equality Constraints

Consider

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & A\mathbf{x} = \mathbf{b}\end{array}$$

- The feasible direction set is $\{\mathbf{d} | A\mathbf{d} = 0\}$.
- The descent direction set is $\{\mathbf{d} | \nabla f(\mathbf{x})^T \mathbf{d} < 0\}$.

The FONC says that at local minimum, there cannot be a solution to both systems (feasible and descent direction) $\Rightarrow \{\mathbf{d} : A\mathbf{d} = 0\} \cap \{\mathbf{d} : \nabla f(\mathbf{x})^T \mathbf{d} < 0\} = \emptyset$

Theorem (Alternative System)

The system $A\mathbf{d} = 0$ and $\nabla f(\mathbf{x})^T \mathbf{d} < 0$ does not have a solution if and only if there exists $\mathbf{y}$ such that

$$A^T \mathbf{y} = \nabla f(\mathbf{x})$$

Nonlinear Optimization with Equality Constraints

Therefore, the first-order necessary condition for

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \end{array} \quad (1)$$

is that there exists $\mathbf{y}$ such that

$$A^T \mathbf{y} = \nabla f(\mathbf{x})$$

then the system $A\mathbf{d} = 0$ and $\nabla f(\mathbf{x})^T \mathbf{d} < 0$ has no soln

Theorem

If $\mathbf{x}^*$ is a local minimum for (1), then there must exist $\mathbf{y}$ such that

$$A^T \mathbf{y} = \nabla f(\mathbf{x}^*)$$

Statement:

The system $\begin{cases} Ad=0 \\ \nabla f(x)^T d < 0 \end{cases}$ has no solution.



$$\exists y \text{ s.t. } A^T y = \nabla f(x)$$

Proof: ① If $\exists y$ s.t. $A^T y = \nabla f(x)$, then the system $\begin{cases} Ad=0 \\ \nabla f(x)^T d < 0 \end{cases}$ has no solution.

multiply d on both sides

$$(A^T y)^T d = (\nabla f(x))^T d$$

$$\Rightarrow (Ad)^T y = \nabla f(x)^T d$$

So $\begin{cases} Ad=0 \\ \nabla f(x)^T d < 0 \end{cases}$ has no soln.

② If $\begin{cases} Ad=0 \\ \nabla f(x)^T d < 0 \end{cases}$ has no soln, then $\exists y: A^T y = \nabla f(x)$

Consider the LP

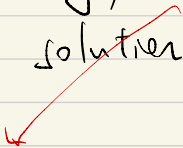
$$(P) \quad \begin{array}{ll} \min. & \nabla f(x)^T d \\ \text{s.t.} & \rightarrow Ad=0 \text{ (y)} \\ & d \text{ free} \end{array}$$

$$(D) \quad \begin{array}{ll} \max. & 0^T y = 0 \\ \text{s.t.} & A^T y = \nabla f(x) \\ & \rightarrow y \text{ free} \end{array}$$

if $\begin{cases} Ad=0 \\ \nabla f(x)^T d < 0 \end{cases}$ has no soln, then (P) cannot be unbounded.

Also since $d=0$ is feasible for (P), so (P) is feasible.
So (P) must have optimal soln.

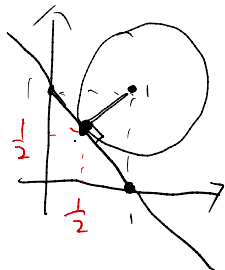
By strong duality, (D) is feasible and has optimal solution.


$$\exists y: A^T y = \nabla f(x)$$

Example

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 - 2 \\ 2x_2 - 2 \end{bmatrix}$$

$$\begin{aligned} \min_{f(x)} & (x_1 - 1)^2 + (x_2 - 1)^2 \\ \text{s.t.} & x_1 + x_2 = 1 \end{aligned}$$



This problem finds the nearest point on the line $x_1 + x_2 = 1$ to the point $(1, 1)$

$$x_1 + x_2 = 1$$

$$\rightarrow Ax = b$$

$$A = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad b = 1$$

At local minimizer x^* , I have $\exists y$ s.t.

$$A^T y = \nabla f(x^*)$$

$$A^T y = \nabla f(x)$$

$\begin{matrix} 2 \times 1 & 1 \times 1 & 2 \times 1 \end{matrix}$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} y = \begin{bmatrix} 2x_1^* - 2 \\ 2x_2^* - 2 \end{bmatrix}$$

$$\begin{cases} 2x_1^* - 2 = y \\ 2x_2^* - 2 = y \end{cases} \Rightarrow 2x_1^* - 2 = 2x_2^* - 2$$

$$\Downarrow$$

$$\begin{cases} x_1^* = x_2^* \\ x_1^* + x_2^* = 1 \end{cases}$$

$$\Downarrow$$

$$x_1^* = x_2^* = \frac{1}{2} \text{ is local min}$$

(global)

Inequality Constraints

Now we consider an inequality constrained problem:

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & A\mathbf{x} \geq \mathbf{b}\end{array} \quad (2)$$

Theorem

If $\mathbf{x}^*$ is a local minimum of (2), then there exists some $\mathbf{y} \geq \mathbf{0}$ satisfying

$$\begin{aligned}\nabla f(\mathbf{x}^*) &= A^T \mathbf{y} \\ y_i \cdot (\mathbf{a}_i^T \mathbf{x}^* - b_i) &= 0, \quad \forall i\end{aligned}$$

where $\mathbf{a}_i^T$ is the i th row of A .

More General Cases

We have discussed unconstrained optimization and constrained optimization with linear equality constraints or linear inequality constraints and derived the (necessary) optimality conditions.

- We want to extend them to more general cases — KKT conditions.
- KKT conditions is the first-order necessary conditions (FONC) for general nonlinear optimization problems.

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Relaxation

$$V_P \stackrel{\text{larger}}{\leq} \min_x \{f(x) : x \in X\}$$


$$\underline{V_Q} \stackrel{\text{smaller}}{\leq} \min_x \{g(x) : x \in Y\}$$

Problem (Q) is a relaxation of (P) if:

- $X \subseteq Y$
- $f(x) \geq g(x) \quad \forall x \in X$

lower bound

$$V_Q \leq V_P$$

Relaxation Properties

- The relaxation of an optimization problem should be easier to solve.
- The optimal value of the relaxation provides a lower bound on the original problem. $V_Q \leq V_P \leq f(x), \forall x \in X$
- If the relaxation is infeasible, then clearly the original problem is also infeasible. $Y = \emptyset, X \subseteq Y \Rightarrow X = \emptyset$
- Suppose only the constraints are relaxed. If ~~a~~ solution to the relaxation is feasible for the original problem, then it must be an optimal solution to the original problem.

$(P) \quad \min. \quad f(x)$
 $\text{s.t. } x \in X$

$(Q) \quad \min. \quad f(x)$
 $\text{s.t. } x \in Y$

$X \subseteq Y$

$x^* \in X \Rightarrow x^* \text{ is optimal for } (P)$

in optimal

Lagrangian Relaxation

General
OPT $\rightarrow$

(P)

Penalty

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq b_i, \quad \forall i \in I \\ & h_j(x) = d_j, \quad \forall j \in J \end{aligned}$$

The Lagrangian function is

$$L(x, \lambda, \mu) = f(x) + \sum_{i \in I} \lambda_i [g_i(x) - b_i] + \sum_{j \in J} \mu_j [h_j(x) - d_j]$$

where $\lambda_i \geq 0$ for all $i \in I$ and μ_j for $j \in J$ are Lagrangian multipliers.
Then the following problem is the Lagrangian relaxation of (P):

$$(Q) : \min_{\substack{x \\ \text{no constraints}}} L(x, \lambda, \mu)$$

$$L(x; \lambda, \mu) = f(x) + \underbrace{\sum_i \lambda_i [g_i(x) - b_i]}_{\substack{\geq 0 \quad \leq 0 \\ \leq 0}} + \underbrace{\sum_j \mu_j [h_j(x) - d_j]}_{\substack{\geq 0 \\ = 0}}$$

$$\leq f(x), \quad \forall x \in X$$

$\min_x L(x; \lambda, \mu)$ is a relaxation problem of (P)

Lagrangian Relaxation

original problem

$$\rightarrow v_P = \min_{\mathbf{x}} \{f(\mathbf{x}) : g_i(\mathbf{x}) \leq b_i, \forall i \in I, h_j(\mathbf{x}) = d_j, \forall j \in J\}$$

For $\lambda_i \geq 0$, let:

$$\mathcal{L}(\lambda, \mu) = \min_{\mathbf{x}} L(\mathbf{x}, \lambda, \mu)$$

$$= \min_{\mathbf{x}} \left\{ f(\mathbf{x}) + \sum_{i \in I} \lambda_i [g_i(\mathbf{x}) - b_i] + \sum_{j \in J} \mu_j [h_j(\mathbf{x}) - d_j] \right\}.$$

Then:

$$\rightarrow \mathcal{L}(\lambda, \mu) \leq v_P \quad \forall \lambda \geq 0.$$

$$d(\lambda, \mu) \leq V_p, \quad \forall \lambda \geq 0$$

$$\text{-----} V_p$$

$$d(10, 1) \text{-----} \lambda=10, \mu=1$$

$$d(50, 10) \text{-----} \lambda=50, \mu=10$$

$$d(100, -50) \text{-----} \lambda=100, \mu=-50$$

We like larger lower bound.

The largest lower bound can be obtained from

$$(D) \quad V_D = \max_{\lambda, \mu} d(\lambda, \mu) \\ \text{s.t.} \quad \lambda \geq 0$$

$$d(\lambda, \mu) \leq V_p, \quad \forall \lambda \geq 0, \forall \mu$$

$$\Rightarrow \max_{\lambda \geq 0, \mu} d(\lambda, \mu) \leq V_p$$

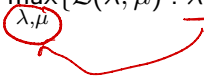
$$\Rightarrow V_D \leq V_p \quad \text{Weak duality}$$

Weak duality

Original Problem:

$$(P) : \quad v_P = \min_x \{f(x) : g_i(x) \leq b_i, \forall i \in I, h_j(x) = d_j, \forall j \in J\}$$

(Lagrangian) Dual Problem:

$$(D) : \quad v_D = \max_{\lambda, \mu} \{ \mathcal{L}(\lambda, \mu) : \lambda \geq 0 \}$$


where

$$\mathcal{L}(\lambda, \mu) = \min_x \left\{ f(x) + \sum_{i \in I} \lambda_i [g_i(x) - b_i] + \sum_{j \in J} \mu_j [h_j(x) - d_j] \right\}.$$

Weak duality: $v_D \leq v_P$

Weak duality always holds!

Example: Two-way Partitioning Problem

$$\begin{array}{ll}\min & x^T W x \\ \text{s.t.} & \underline{x_i^2 = 1, \forall i = 1, \dots, n} \quad (\mu_i)\end{array}$$


- Feasible region $\{-1, 1\}^n$ contains 2^n discrete points
- Interpretation: partition in two sets, $x_i \in \{-1, 1\}$ is an assignment for item i .

Lagrangian:

$$L(X, \mu) = X^T W X + \sum_{i=1}^n \mu_i (X_i^2 - 1)$$

Lagrangian Relaxation

$$\min. L(X, \mu) = X^T W X + \sum_{i=1}^n \mu_i (X_i^2 - 1)$$

$$= X^T W X + \underbrace{\sum_{i=1}^n \mu_i X_i^2} - \sum_{i=1}^n \mu_i$$

$$\sum_i \mu_i X_i^2$$

$$= [X_1 \dots X_n] \begin{bmatrix} \mu_1 & & \\ & \ddots & \\ 0 & & \mu_n \end{bmatrix} \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} = X^T W X + X^T \text{diag}(\mu) X - 1^T \mu$$

$$\min. X^T (W + \text{diag}(\mu)) X - 1^T \mu$$

$$= X^T \text{diag}(\mu) X = \begin{cases} -1^T \mu & \text{if } W + \text{diag}(\mu) \preceq 0 \\ -\infty & \text{o.w.} \end{cases}$$

$$\min. aX^2 - b$$

$$= \begin{cases} -b & \text{if } a \geq 0 \\ -\infty & \text{if } a < 0 \end{cases}$$

Lagrangian dual.

$$\max_{\mu} -1^T \mu$$

$$\text{s.t. } W + \text{diag}(\mu) \preceq 0$$

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General Optimality Conditions

Now, We consider the following nonlinear program:

$$\begin{array}{ll} \min_{x \in \mathbb{R}^n} & f(x) \\ \text{subject to} & g_i(x) \leq 0, \quad \forall i = 1, \dots, m, \\ & h_j(x) = 0, \quad \forall j = 1, \dots, p. \end{array}$$

Handwritten annotations: λ_i (red arrow pointing to $g_i(x) \leq 0$), μ_j (red arrow pointing to $h_j(x) = 0$), $g_i(x) < 0$ and $g_i(x) = 0$ (handwritten branches from $g_i(x) \leq 0$).

We will also work with the following index sets:

Definition: Active and Inactive Set

At a point $x \in X$, the set $\mathcal{A}(x) := \{i : \underline{g_i(x)} = 0\}$ denotes the set of **active constraints**. The set of **inactive constraints** is then given by $\mathcal{I}(x) := \{i : \underline{g_i(x)} < 0\}$.

Karush-Kuhn-Tucker Conditions

If x is a local minimizer and if a constraint qualification (CQ) holds, then there exist λ and μ such that the following **Karush-Kuhn-Tucker (KKT) Conditions** hold:

- ① Main/Stationarity Condition

$$0 \in \partial_x L(x, \lambda, \mu)$$

$$\nabla_{\underline{x}} L(x, \lambda, \mu) = \nabla f(x) + \sum_{i=1}^m \lambda_i \nabla g_i(x) + \sum_{j=1}^p \mu_j \nabla h_j(x) = 0$$

- ② Complementary Slackness

$$\lambda_i \cdot g_i(x) = 0 \quad \forall i = 1, \dots, m$$

- ③ Primal Feasibility

$$g_i(x) \leq 0, \quad h_j(x) = 0 \quad \forall i = 1, \dots, m, \quad \forall j = 1, \dots, p$$

- ④ Dual Feasibility

$$\lambda_i \geq 0 \quad \forall i = 1, \dots, m$$

Constraint Qualifications

We require the collection of gradients

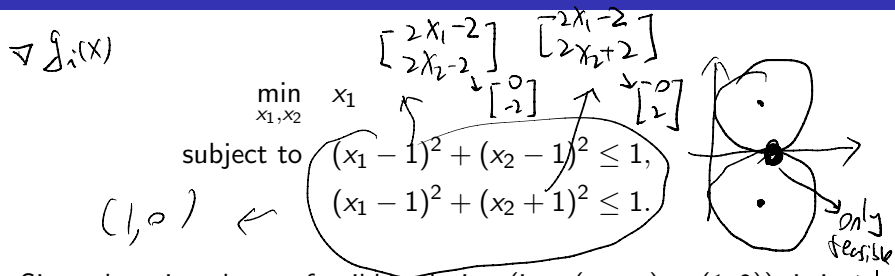
$$\{\nabla g_i(x) : i \in \mathcal{A}(x)\} \cup \{\nabla h_j(x) : j = 1, \dots, p\} \quad (\text{CQ})$$

Handwritten note: $\nabla g_i(x) \neq 0$

to be **linearly independent** or to have full rank.

- This condition is a **constraint qualification (CQ)** and is called **Linear Independence Constraint Qualification (LICQ)**.
- A feasible point x satisfying the LICQ is called **regular**.
- There are more CQs: ACQ, GCQ, MFCQ, PLICQ, Slater's condition.
- If the local minimizer x^* does not hold a CQ, then KKT condition may not be necessary.

Failure of The KKT Conditions in the Absence of CQ



- ▶ Since there is only one feasible solution (i.e., $(x_1, x_2) = (1, 0)$), it is *so* automatically optimal.
- ▶ However, this point does not satisfy the KKT ^{max} conditions (exercise).
- ▶ This example says if a local min does not hold certain CQ, the KKT conditions may not be necessary at this point.
- ▶ This means that points satisfying KKT conditions **may not** contain all the local minimizers, since some of them may not be regular.

Remarks on KKT Conditions

- KKT conditions is **first-order necessary conditions (FONC)** for general constrained optimization problems.
- KKT conditions unify all formerly studied FONC.
- A (feasible) point (often the primal and dual variables (x, λ, μ) together) satisfying the KKT conditions is called a **KKT point**, regardless whether it satisfies CQ or not. We may also call x a KKT point for simplicity.
- **KKT points** are candidates for local optimal solutions – just like stationary points.

Observations:

$$\lambda_i g_i(x) = 0$$

- The Lagrange multipliers (dual variables) $\lambda_i = 0, i \in \mathcal{I}(x)$.
- If a local minimizer x satisfies **LICQ**, then the Lagrange multipliers (dual variables) λ, μ solved by the KKT conditions must be **unique**.

math,

$$\nabla L(x, \lambda, \mu) = \nabla f(x) + \sum_i \lambda_i \nabla g_i(x) + \sum_j \mu_j \nabla h_j(x) = 0$$

$$= \nabla f(x) + \underbrace{[\lambda_1, \dots, \lambda_m, \mu_1, \dots, \mu_p]}_{y^T} \begin{bmatrix} \nabla g_1(x) \\ \vdots \\ \nabla g_m(x) \\ \nabla h_1(x) \\ \vdots \\ \nabla h_p(x) \end{bmatrix} = 0$$

A

$$\nabla f(x) + A^T y = 0$$

A has full rank $\Rightarrow y$ is unique.

Strategy for Applying KKT Conditions

We can summarize some strategies before studying some examples:

General Strategy:

- Check LICQ (if required).
- Derive KKT conditions.
- Start by discussing different easy cases via the complementarity conditions (set multiplier or constraints to 0), then find all KKT points.

$$\lambda_i g_i(x) = 0 \quad \begin{cases} \lambda_i = 0 \\ g_i(x) = 0 \end{cases}$$

Additional Information:

- If the LICQ holds, then λ and μ are always unique.
- If CQ holds, then the global optimizer (assume it exists) must be a KKT point. If further there is a unique KKT point, then this point must be global minimizer.
- (Later) If the problem is convex, then one does not need to care about CQ, any KKT point will sufficiently be a global minimizer.

OPT

- (a) Write down all KKT conditions,
- (b) Find all KKT pts.
- (c) Identify an optimal soln.

Know have global optimal soln and
only have 1 KKT point,



KKT pt is global optimal

Example 1

Consider the problem:

$$\begin{aligned} & \text{minimize} && 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ & \text{subject to} && x_1^2 + x_2^2 \leq 5 \quad (\lambda_1) \\ & && 3x_1 + x_2 \geq 3 \quad (\lambda_2) \end{aligned}$$

Find the KKT conditions.

$$\begin{aligned} \mathcal{L}(x, \lambda_1, \lambda_2) = & 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ & + \lambda_1 (x_1^2 + x_2^2 - 5) \\ & + \lambda_2 (3 - 3x_1 - x_2) \end{aligned} \quad \lambda_1, \lambda_2 \geq 0$$

KKT Conditions $a+b+c+d$

Main:

$$\nabla_x L(x, \lambda_1, \lambda_2) = \begin{bmatrix} \frac{\partial L}{\partial x_1} \\ \frac{\partial L}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 4x_1 + 2x_2 - 10 + 2\lambda_1 x_1 - 3\lambda_2 \\ 2x_1 + 2x_2 - 10 + 2\lambda_1 x_2 - \lambda_2 \end{bmatrix}$$

$$(a) \begin{cases} 4x_1 + 2x_2 - 10 + 2\lambda_1 x_1 - 3\lambda_2 = 0 \\ 2x_1 + 2x_2 - 10 + 2\lambda_1 x_2 - \lambda_2 = 0 \end{cases}$$

Complementary Slackness

$$(b) \begin{cases} \lambda_1 (x_1^2 + x_2^2 - 5) = 0 \\ \lambda_2 (3 - 3x_1 - x_2) = 0 \end{cases}$$

Primal feasible

$$(c) \begin{cases} x_1^2 + x_2^2 \leq 5 \\ 3x_1 + x_2 \geq 3 \end{cases}$$

Dual feasible

$$(d) \lambda_1, \lambda_2 \geq 0$$

Example 2

Find the KKT conditions for

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \quad (\mu) \\ & \mathbf{C} \mathbf{x} \geq \mathbf{d} \quad (\lambda_1) \\ & \underline{\mathbf{x}} \geq \mathbf{0} \quad (\lambda_2) \end{aligned}$$

$$\begin{aligned} L(\mathbf{x}, \lambda, \mu) = & \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x} + \mu^T (\mathbf{A} \mathbf{x} - \mathbf{b}) \\ & + \lambda_1^T (\mathbf{d} - \mathbf{C} \mathbf{x}) \\ & - \lambda_2^T \mathbf{x} \quad , \end{aligned}$$

$\lambda_1, \lambda_2 \geq 0$

KKT Conditions: $a+b+c+d$

Main

$$\nabla_x L(x, \lambda, \mu) = 2Qx - c + A^T \mu - C^T \lambda_1 - \lambda_2$$

(a) $2Qx - c + A^T \mu - C^T \lambda_1 - \lambda_2 = 0$

Complementary Slackness

(b)
$$\begin{cases} \lambda_{1,i} (d - Cx)_i = 0, \forall i \\ \lambda_{2,i} x_i = 0, \forall i \end{cases}$$

Primal feasible

(c) $Ax = b, Cx \geq d, x \geq 0$

Dual feasible

(d) $\lambda_1, \lambda_2 \geq 0$

Example 3: Cylinder Volume

We want to build a cylinder with the maximum volume, with its surface area no larger than C .

- Decision variables: r (the radius of the base) and h (height).
- Then the optimization problem is:



maximize r, h

subject to $2\pi r^2 + 2\pi rh \leq C \quad (\lambda_1)$

$r, h \geq 0 \quad (\lambda_2, \lambda_3)$

$$L(r, h, \lambda) = -\pi r^2 h + \lambda_1 (2\pi r^2 + 2\pi rh - C) - r\lambda_2 - h\lambda_3$$

$\lambda_1, \lambda_2, \lambda_3 \geq 0$

Handwritten notes: A blue arrow points from the $\pi r^2 h$ term in the constraint to the $-\pi r^2 h$ term in the objective function. Another blue arrow points from the r, h in the objective function to the r, h in the constraint. The r, h in the objective function are circled in red.

KKT conditions $a+b+c+d$

Main:

$$(a) \begin{cases} \frac{dL(r, h, \lambda)}{dr} = -2\pi rh + 4\lambda_1 \pi r + 2\lambda_1 \pi h - \lambda_2 = 0 & (1) \\ \frac{dL(r, h, \lambda)}{dh} = -\pi r^2 - 2\lambda_1 \pi r - \lambda_3 = 0 & (2) \end{cases}$$

Complementary Slackness

$$(b) \begin{cases} \lambda_1 \cdot (2\pi r^2 + 2\pi rh - C) = 0 & (3) \\ \lambda_2 \cdot r = 0 & \lambda_3 \cdot h = 0 & (4) \end{cases}$$

Primal & dual feasible

$$(c) \quad 2\pi r^2 + 2\pi rh \leq C, \quad r \geq 0, \quad h \geq 0$$

$$(d) \quad \lambda_1, \lambda_2, \lambda_3 \geq 0$$

$$r > 0, h > 0 \xrightarrow{(4)} \lambda_2 = 0, \lambda_3 = 0$$

From (3), Case I: $\lambda_1 = 0$ Plug in (2)

$$-\pi r^2 = 0 \Rightarrow r = 0 \quad \text{Impossible!}$$

$$\text{Case II: } \begin{cases} 2\pi r^2 + 2\pi rh = C & (3') \\ -2\pi rh + 4\lambda_1 \pi r + 2\lambda_1 \pi h = 0 & (1) \\ -\pi r^2 + 2\lambda_1 \pi r = 0 & (2) \end{cases}$$

$$\text{From (2)} \quad 2\lambda_1 r = r^2 \Rightarrow r = 2\lambda_1$$

Plug in (1)

$$-2\pi(2\lambda_1)h + 4\lambda_1\pi \cdot (2\lambda_1) + 2\lambda_1\pi h = 0$$

$$\Rightarrow -4h + 8\lambda_1 + 2h = 0$$

$$\Rightarrow h = 4\lambda_1$$

plug in (3')

$$2\pi(2\lambda_1)^2 + 2\pi(2\lambda_1)4\lambda_1 = C$$

$$\Rightarrow 8\pi\lambda_1^2 + 16\pi\lambda_1^2 = C$$

$$\Rightarrow 24\pi\lambda_1^2 = C$$

$$\Rightarrow \lambda_1 = \sqrt{\frac{C}{24\pi}}$$

The
only
KKT
Pt

$$\left\{ \begin{array}{l} r^* = 2\lambda_1 = 2\sqrt{\frac{C}{24\pi}} \\ h^* = 4\lambda_1 = 4\sqrt{\frac{C}{24\pi}} \end{array} \right.$$

$\Rightarrow$ global
 $\nearrow$ optimal

Example 4: Power Allocation

We have a collection of n communication channels and we need to decide how much power to allocate to each of them

- The capacity (communication rate) of channel i is $\log(\alpha_i + x_i)$ with a given $\alpha_i > 0$ and when x_i is allocated to it, and we have a budget constraint $\mathbf{e}^T \mathbf{x} = 1$, $\mathbf{x} \geq 0$

The optimization problem is:

$$\begin{aligned} & \text{maximize}_{\mathbf{x}} \quad \sum_{i=1}^n \log(\alpha_i + x_i) \quad \hookrightarrow \text{min. } -\sum_i \log(\alpha_i + x_i) \\ & \text{subject to} \quad \sum_{i=1}^n x_i = 1 \quad (\mu) \\ & \quad \quad \quad \mathbf{x} \geq 0 \quad (\lambda) \end{aligned}$$

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = -\sum_{i=1}^n \log(\alpha_i + x_i) - \underbrace{\sum_i \lambda_i x_i}_{\lambda^T \mathbf{x}} + \mu \left(\sum_{i=1}^n x_i - 1 \right)$$

KKT conditions: $a+b+c+d$

Main

$$\nabla_x L(x, \lambda, \mu) = \begin{bmatrix} \frac{\partial L}{\partial x_1} \\ \vdots \\ \frac{\partial L}{\partial x_n} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{a_1 + x_1} - \lambda_1 + \mu \\ \vdots \\ -\frac{1}{a_n + x_n} - \lambda_n + \mu \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\Rightarrow \text{(a)} \quad -\frac{1}{a_i + x_i} - \lambda_i + \mu = 0, \quad \forall i=1, \dots, n$$

Complementary Slackness

$$\text{(b)} \quad \lambda_i x_i = 0 \quad \forall i$$

$$\begin{cases} \lambda_i = 0 & \textcircled{1} \\ x_i = 0 & \textcircled{2} \end{cases}$$

Primal & dual feasible

$$\text{(c)} \quad \sum_i x_i = 1, \quad x \geq 0$$

$$\text{(d)} \quad \lambda \geq 0$$

① If $\lambda_i = 0$

$$(a) \Rightarrow -\frac{1}{a_i + x_i} + \mu = 0$$

$$x_i = \frac{1}{\mu} - a_i$$

require $x_i \geq 0 \Rightarrow \frac{1}{\mu} - a_i \geq 0 \Rightarrow \underline{\mu \leq \frac{1}{a_i}}$

② If $x_i = 0$

$$(a) \Rightarrow -\frac{1}{a_i} - \lambda_i + \mu = 0$$

$$\lambda_i = \mu - \frac{1}{a_i}$$

require $\lambda_i \geq 0 \Rightarrow \mu - \frac{1}{a_i} \geq 0 \Rightarrow \underline{\mu \geq \frac{1}{a_i}}$

$$x_i^* = \begin{cases} 0 & \text{if } \mu \geq \frac{1}{a_i} \\ \frac{1}{\mu} - a_i & \text{if } \mu < \frac{1}{a_i} \end{cases}$$

$$= \max \{0, \frac{1}{\mu} - a_i\}$$

Use $\sum_i x_i^* = \sum_i \max \{0, \frac{1}{\mu} - a_i\} = 1$
to find out μ .